

Jashore University of Science and Technology

Department of Physics

Bachelor of Science with Honours in Physics

2nd semester of 3rd year (2024–2025)

Course code: PHY 3205

Course title: Solid State Physics I

Assignment no.: 1

Date: 06 May 2026

Deadline for submission: 14 May 2026, 10:00 PM

1. Classify crystal defects and explain their physical origin. How do point, line, and plane defects impact on material properties? How does temperature influence defect concentration?
2. Explain different types of point defects (vacancies, interstitials, substitutional defects). Compare Schottky and Frenkel defects in ionic solids.
3. Explain the structure of edge dislocation and screw dislocation. Compare edge and screw dislocations in terms of geometry.
4. What are plane defects? Discuss different types such as grain boundaries, twin boundaries, and stacking faults.
5. What are color centers? Explain F-centers with an energy-level diagram. How do color centers form in ionic crystals? Discuss their optical consequences.